

EDGELESS CROWDSALE WHITEPAPER

December, 2017

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Restatement and New Wording

This is a new wording of Edgeless Casino Crowdsale Whitepaper. This Whitepaper has been updated taking into consideration the developments in cryptocurrency regulation and progress in the implementation of the Edgeless Casino project. This new wording of Edgeless Casino Crowdsale Whitepaper entirely restates and replaces the Edgeless Casino Crowdsale Whitepaper published in early 2017, which will automatically become null and void as of publication of this new wording of Edgeless Casino Crowdsale Whitepaper.

Executive Summary

The online gambling industry was valued at USD 45.86 billion in 2016. However, it has two major problems: the **price** of the gamble and **transparency**. The price of the individual gamble comes from user-unfriendly rules: a large house edge of 1% - 15% (ref. Expn Section nr. 1), money deposit/withdrawal fees and 2 - 3 days withdrawal delays. In addition, there is no way of knowing what is happening inside the casino's servers and how the mechanisms are programmed. It can be easily rigged. Players need to rely on trusting the casino, which is most likely not the best option in such a profit-driven industry.

In 2008, Bitcoin and cryptocurrencies solved the gambling price problem. Cryptocurrency casinos are cost effective and can offer a 0.01% - 2% house edge, with instant money transactions without or little fees. However, the **transparency** problem is still there. Casinos can easily rig the game or hide profitability from the investors.

Using Ethereum smart contracts (ref. Expn Section Nr. 1.5), we can have a low price per gamble and solve the transparency problem. Random numbers and payouts can be ruled by smart contracts which are fully public and implemented on an Ethereum blockchain. Now players can see everything: transactions, sums, bets, the reasons why transactions are moving and, most importantly, the mechanism of randomness. If the casino tried to rig the odds, it would be immediately spotted.

Our team brings the first transparent and selected zero edge casino games to the market - **Decentralised Edgeless Casino**. At this point in time, there are no casinos like this in the world.

Our casino will offer a 0% edge Blackjack as well as other games like Dice and Sports Betting. All games will be based on the Ethereum smart contract and will ensure transparency.

The business will sustain itself in two ways:

1. From the players' imperfect play. Blackjack and Poker are games of skill and luck. If players make perfect decisions, they will reach a 0% edge. on average, however, a

player makes mistakes, so it will give a -0.83% edge to the casino and this 0.83% will be earned from the money wagered.

2. Sports Betting is about connecting players who are betting against each other and taking the facilitation fee to manage the process. The fee depends on the game types, the number of participants, bonuses and more. It is dynamic, floating in the range of 0% - 10%. Approximately 4% is earned from the money wagered.

Edgeless project:

- Edgeless Casino Project unites players that want a fairer and more transparent online gaming industry and want to support its creation.
- Edgeless Casino is fully transparent since it is based on Ethereum Smart contracts.
 - Edgeless Casino players will trust that the casino will not cheat them.
 - Bankroll of Edgeless Casino will be formed by Edgeless Casino unless the Edgeless Casino decides otherwise.
 - EDG Token holders can access the Edgeless Casino and play available online games.
- Edgeless Casino is the first casino with 0% edge based on the Ethereum blockchain.
- Edgeless Casino will have strong competitive advantages (0% edge, transparency) to increase its market share in order to become a sustainable business.
- Edgeless Casino can provide many gambling games (BlackJack, Poker, Dice, Sports Betting).

Let's decentralise and solve the main online gambling problems!

	Traditional Online Casino	Bitcoin Online Casino	Edgeless Online Casino (Based on Ethereum Smart Contracts)
House Edge	1% - 15%	1% - 0.1%	0%
Free money withdrawals and deposits	x	Yes	Yes
No delay to money withdrawals and deposits	x	Yes	Yes

Fully transparent random number fairness check	x	x	Yes
Transparency guided by Ethereum smart contract	x	x	Yes

1. Price and Transparency Problems of Online Gambling

As stated before, in 2016 the online gambling industry was valued at USD 48.6 billion and has shown steady growth each year. Most of a casino's profits come from the "house edge". The House Edge is a term used to describe the mathematical advantage that the gambling game (and therefore the commercial gambling venue) has over you as you play. This advantage results in an assured percentage returned to the house over the duration of playing time, and for you, it is an assured percentage loss of what you bet.

Imagine that you are playing roulette and you bet \$100 on the colour red. As roulette has a number ZERO coloured in green which is neither red or black, the house gets an advantage of 2.7% that is calculated thus:

$$100 : (36 + 1) = 2.7\%$$

So if the house has an edge, you ultimately lose every time. If you bet \$10, your expected loss is \$0.27. In one hour, you can spin approximately 20 - 25 times. If every time you bet \$10, then in that hour you are expected to lose \$5.4 - \$6.75.

This loss can be calculated using the expected value (EV) formula:

$$EV = (\text{prob.win})(\text{gain}) - (\text{prob.lose})(\text{gain})$$

Let's say you are betting \$10 on red which gives you an expected \$20 win and \$10 is your expected loss. Roulette has 18 black fields, 18 red fields and 1 green zero. When you put all the variables in the equations, this is what you get:

$$EV = \$20(18/37) + \$0(19/37) = \$9.73 + \$0 = \$9.73$$

If you bet \$10 on a roulette red, at the end of the game then you are expected to have \$9.73. And \$0.27 is lost due to the house edge.

This means that for each \$10 that you bet, you are giving away 27 cents to the house. Of course in real life, numbers do not hit exactly like that and we may have a bigger range of wins and losses.

The house edge is different for each game. [Here you can find a list of games and house edges.](#)

Overall we have two major problems in the online gambling industry: the price of the gamble and house transparency. As this industry is growing rapidly, these problems become ever more relevant.

1.1. Rise of the Cryptocurrency Reduce the Gamble Price

One of the reasons why online casinos have a deposit fee, withdrawal fees and extended withdrawal times are to do with problems related to traditional forms of currency such as USD and EUR. To move traditional currency from one account to the other is expensive. It has to go through many third parties and regulators which take their bit out of every transaction. In 2008, cryptocurrency Bitcoin came to life and started changing a lot of areas of our lives including the online gambling industry. By providing instant and low cost currency transactions, Bitcoin became a game changer for online casinos. Some rooms offer a house edge that is lower than 1%, which is more of a fair bet. This significantly improves the quality of the gamble as a whole.

That's why cryptocurrency-based online casinos are on the rise. Right now, we have more than 100 rooms based on a cryptocurrency which accounts for 10 billion bets and is growing each year. In fact, 60% of all Bitcoin transactions are gambling-related. However, one major problem has not been solved yet: **transparency**. Even after the cost of the gamble has been reduced significantly, we still have many rooms rigging the odds and stealing from customers. Famous cases of Bitcoin casinos rigging the odds can be founded [here](#) and [here](#).

So, one major problem is left: transparency.

1.2. Fairness Check's Problems and Limitations

1.2.1 Random Number Generator

1.2.1.1. Explanation of Fairness Check

Online casinos create a random number using a combination of casino and client numbers. The casino assigns their own number automatically and the visitor can do it manually or let the casino decide for them. The two numbers are blended to provide a random outcome. The idea behind it is that the casino can rig their own number, but they would not be able to rig the client's number and therefore the outcome will be random. Therefore a casino would not be able to cheat.

Oversimplified example: casino is assigned a number (also known as a seed), visitor assigns the other number (seed). The visitor can set it manually before every game or let the casino decide. Both numbers are blended into one number and using a formula (it's unknown to the player and determined by the casino otherwise the player could predict the final outcome) - **the random number is generated**. After the game, visitors can use a special key (hash, information access code/label/tag) to access the information related to that game and see what seeds were blended and which number was generated as the result.

1.2.1.2. Loops in Fairness Check

- 1) The casino can easily influence the client's seed by assigning it themselves (that's the case in most casinos, because an average player is not going to type and give a random number before every gamble round - even if they have this option)
- 2) The client entered the seed once and is not changing it. The casino can identify it and use that information to adjust their own seed.
- 3) The casino can delay creating their own seed and provide it to the 'blender' once the visitor has sent theirs. Then the casino can adjust their own seed to the client's.

Even if client sees that the 'randomness' of the casino is on the right track, the casino still has the ability to freeze a player's account without reason and steal the deposit, since these actions depend on humans' decisions but not on non-changeable fully transparent and decentralised smart contracts. There are so many ways of cheating that it is clear that a new generation of online casinos is needed.

1.2.2 Bitcoin Casino Can Also Drain Investors Profits

A lot of cryptocurrency casinos offer investment opportunities for people. They can invest into the casino's bankroll and get a share of a casino's profits. All cryptocurrency transactions are public on the blockchain so it is easy to see the exact profits every day. That's why this investment looks really attractive. However, there is a way that casinos can drain profits from their wallet and pay less to investors.

For example, a casino shares 50% of its profit to its investors. In 2016, they made \$1,000,000\$ profit and \$500,000 should therefore go to the investors. However, the casino wants to cheat and pay less. They can easily drain money from the casino's wallet into a different wallet which belongs to them and no-one would even suspect that it has happened. They would do this by:

1. Casino owner enters his own online casino with a new account and different wallet.
2. Casino edits code in casino server and riggs odds ONLY for that new account.
3. Casino owner wins against casino and leaves.
4. The casino pays less profit share to its investors, because they think casino just had a loss and someone got lucky. But they have no idea that the lucky person is the owner of the casino.

Blockchain can see all transactions and the sums transferred. However they cannot see the reasons why transactions are moving or what is happening inside the casino's server. This scheme can be done in low sums and systematically over the year. At the end of the year, investors would see the total casino's profit and calculate the EV based on their edge. E.g. casino's profits are 10% less than expected, however, no one would suspect that the owner took it. And 10% from half a million is \$50,000. That is one way that casinos can lower their public profits and steal money as a result.

1.3. Edgeless Casino Solution

This is why we will be developing Edgeless - a next generation casino. Edgeless casino will be built on the Ethereum blockchain. All of the key operations will be regulated by smart contracts, giving users a fully transparent experience.

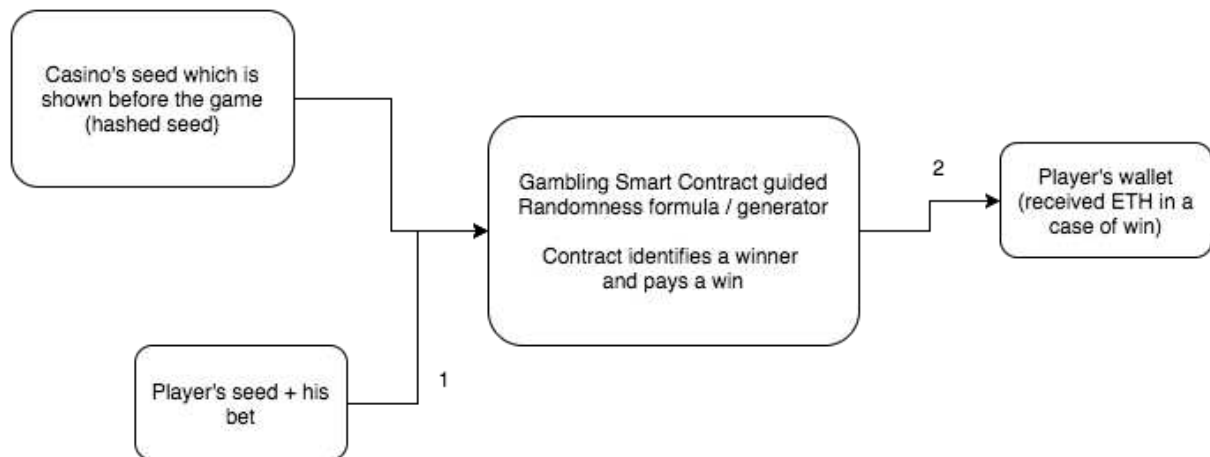
We will be using the client's seed and our own seed to generate randomness. Users will know our hashed (SHA512) seed before betting and before generating their own client seeds. Because of that there is no way for us to alter the results. After bet is made, client will be able to see our unhashed seed on Ethereum Blockchain and verify the randomness. Moreover, clients will have two options for getting random seeds. The first one is to type it in manually before every game; this would eliminate any ability to cheat the clients. The second one, which will be used by default, is to generate different client seeds automatically for every game by using an API integration with random.org. Smart contracts will be publicly available on the blockchain for everyone to examine and to test its randomness procedure.

1.4. Edgeless Based on Ethereum Smart Contracts - Full Transparency

In 2015, the creation of the Ethereum smart contract is another major game changer for online gambling. Right now, we can solve the transparency issue once and for all. Using a smart contract, there is no way that casinos can rig the odds even if they wanted to. That means in 2017-2018, we can have "THE PERFECT ONLINE CASINO" which can offer 0% house edge and 100% transparency. That's a next generation gambling product which is the best in a current market and does not exist. Yet.

The real power of smart contracts is that if they are properly implemented on Ethereum Blockchain, they will act each and every time how they are supposed to. That is truly trust-flawless technology that eventually all online casinos will be using. Also, it is near impossible to censor smart contracts, because they are placed in a decentralized infrastructure. All Ethereum smart contracts will be publicly available for everyone to see and examine in publicly accessible websites.

In the same trustworthy way that our casino will be operated, if a user bets on a particular game, they can be 100% sure that their result will be truly random and that they will get their win transferred directly into their wallet. The casino will not be able to suspend the transaction or redirect it to another wallet.



1. Player's bet transaction to the Casino Smart Contract Wallet (ETH currency related)
2. Smart contract pays a win (ETH currency related)

Graph - Edgeless randomness and fairness mechanism based on Ethereum. Player wallet will receive Edgeless token in case of a win and all games will be played with Edgeless token.

2. How Edgeless Casino Can be a Profitable Business Offering 0% House Edge?

2.1. Player Mistakes and Imperfect Play

Most casino games do not require any skill to play or change the odds like roulette, dice, and slots. However, there are a few games which combine luck and skill. By providing more options - blackjack, video poker - players can influence house edge. That means that the

house edge depends on the skill level of the player. Blackjack is the game which has the lowest casino edge (0.6%) if played perfectly. The average player, however, does not play the 'perfect game' and they make mistakes. Black Jack provides a variety of decisions to the player and based on that, they can increase or decrease their chance of winning. On average, a player makes mistakes which are equivalent of 1.43%. If we offer a 0% house edge game, mistakes count for a 0.83% house edge. That means on every \$100 bet, an average player gives 83 cents to the house.

Take a look at full report and analysis of blackjack house edge and player mistakes:[here](#)

2.2. Gambler's' Ruin and Variance

Gambler's ruin adds an extra profit to the casino. A gambler with finite wealth playing a fair game (both sides have equal chance of winning - 0% house edge) would lose against an opponent with infinite wealth. It's quite simple; a larger bankroll can absorb higher levels of variance. And small bankrolls can't. That means a small player would bust more even if the casino has a 0% edge.

Let's take an example:

"Assume that the player makes a deposit of \$1000 to an online casino, and wants to bet till he reaches \$5000 that has been wagered. If the player is willing to play through 500 hands, then his average bet size would be $\$5000/500 = \10 . The number of betting units would be $\$1000/\$10 = 100$. The table shows the risk of ruin is 0.01% for 102 units, so would be just over 0.01% for 100. Perhaps this is too conservative, so the player considers playing 200 hands. The bet size is now $\$5000/200 = \25 . The number of units is $\$1000/\$25 = 40$. Interpolating the table shows the risk of ruin would be 1.5%."

Number of Hands to Play										
RISK OF RUIN	100	200	300	400	500	600	700	800	900	
50%	7	11	14	16	18	20	22	24	25	
40%	9	14	17	20	23	25	27	29	31	
30%	12	17	21	25	28	31	33	36	38	
20%	15	21	26	31	34	38	41	44	47	
10%	19	27	34	39	44	48	53	57	60	
5%	22	32	40	46	52	58	62	67	71	
4%	23	34	42	49	55	60	65	70	75	
3%	25	36	44	51	58	64	69	74	79	
2%	27	38	47	55	62	68	74	79	84	
1%	29	42	52	61	68	75	82	88	93	
0.5%	32	46	57	66	74	82	89	95	101	
0.25%	35	50	61	71	80	88	96	102	109	
0.1%	38	54	67	77	87	95	104	111	118	
0.01%	45	64	79	91	102	112	122	131	139	

Source: <http://wizardofodds.com/games/blackjack/appendix/12/>

More info: <http://wizardofodds.com/games/blackjack/appendix/4/>

At this stage, we cannot estimate the profits coming from the gambler's ruin effect. It strongly depends on the type of players gambling in the casino and their bankroll management skills.

2.3. How will We Reduce an Edge to 0%?

Casino has a blackjack edge because they have best decision making position - the last one. That means that players can bust before the house even draws a card. To compensate for this edge, the house gives additional bonuses to the player such as 1 to 3 payoff in the case of blackjack, and the ability to double bet if casino has a bad card. That's exactly what we will do - increase these additional bonuses by reducing house edge to 0%. Actually such game models already exist - Voyages Casino has one. However, they charge 10% for money withdraws and there is no way of knowing if they are rigging the odds or not.

You can play a 0% edge blackjack demo here: <https://betvoyager.com/games/equal-odds/>

2.4. Games Provided by Edgeless Casino

Based on results of the Crowdsale, these are the following games which Edgeless Casino will offer.

0% Edge Blackjack - the payout structure will be changed in favour of the player, so the house edge would be reduced to ZERO. Here, you can play a demo of zero edge blackjack: <https://betvoyager.com/games/equal-odds/>

Dice. The game rules result in a 0.05% edge. The concept is similar to most bitcoin dice casinos, however the payout structure will be fairer. If you bet 1BTC on 50%, most Bitcoin casinos will offer 1.98x payout. We will offer a 1.995x fair bet payout.

Sports Betting - it's not really a traditional casino gambling game, but in order to attract a wider scope of people, we want to offer more games.

In the future, we will expand on the games offered by our casino.

3. Participatory function of EDG tokens

3.1. The function of EDG tokens

Edgeless Casino Project aims at bringing more transparency into the gambling industry and therefore changing it from the roots. To reach this aim we want to form a community which would believe and support the idea.

The EDG Token will be used to play in the Edgeless Casino available games.

EDG Token will not offer any right to profits, dividends or other right to revenue of the Edgeless Casino. EDG Token is purely utility token to power Edgeless infrastructure.

After the launch Edgeless Casino will not sell any EDG tokens or offer exchange services to other cryptocurrencies or FIAT. Third party providers could offer such service.

4. Edgeless Casino Road map

4.1. Market Potential

The more community members and players the Edgeless Casino have, the bigger bank-roll can be formed and won on the Edgeless Casino.

The online gambling casino industry is valued at USD 45.86 billion and is expected to grow up to USD 56 billion in 2018. Every year, more and more market share of that is going into cryptocurrency gambling solutions. The industry has two major problems: transparency and price of the online gamble.

- At this point, online casinos have full power to cheat without being spotted.
- At this point, players need to pay deposit/ withdrawal fees, wait for payouts and gamble on 1% - 15% house edge.

It all can be changed with the Edgeless Ethereum-based Casino.

Comparison table shows the main Edgeless casino based on Ethereum competitive advantages for attracting player against online and bitcoin Casinos

	Traditional Online Casino	Bitcoin Online Casino	Edgeless Online Casino (Based on Ethereum Smart Contracts)
House Edge	1% - 15%	1% - 0.1%	0%
Free money withdrawals and deposits	x	Yes	Yes
No delay to money withdrawals and deposits	x	Yes	Yes
Fully transparent random number fairness check	x	x	Yes

Transparency guided by Ethereum smart contract	x	x	Yes
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4.2. Marketing Plan

We do believe that having the best product/best offer is the key. Our team is planning to offer perfect gambling conditions: 0% house edge on selected games and ultimate transparency (Ethereum based smart contracts). Right now, there is no product like that in the market - our casino will be the first one.

Our main marketing messages:

- Perfect gambling experience with 0% house edge on selected games
- Fully transparent casino based on Ethereum
- Fast gaming like on Bitcoin casinos even though Edgeless is based on Ethereum blockchain
- 0% Edge BlackJack

In the first stage, we will target players who are already familiar with cryptocurrencies. Organically we expect to attract a larger proportion of gamers from traditional casinos. Edgeless casino will be regulated with specific license and will perform necessary KYC (know your customer) procedures as required by the license requirements.

After the successful Crowdsale, we will put a lot of focus into product development, legal procedures and capturing market share from other cryptocurrency casinos. The development process, business operations, and legal procedures will take 6-12 months. After all of the preparation procedures of Edgeless Casino's development and licensing, we will implement necessary marketing strategies. All marketing efforts will be put in to attract people from cryptocurrency , in turn boosting the overall popularity of cryptocurrencies and fair gambling. Having a 0% Edge Casino on selected games is a strong advantage to attract users to Edgeless Casino, where they can find many other gambling and betting games. One of these will be Sports betting, which will have a 4% commision fee.

5. Crowdsale

The Crowdsale of Edgeless and the corresponding token creation process will be organised around the smart contracts running on Ethereum. Participants willing to support the development of the Edgeless Casino Project can do so by sending Ethereum to the designated address. By doing so, they are purchasing Edgeless Tokens (EDG) at the rate of 1 000 EDG per 1 ETH which are sent instantly to their wallet.

Crowdsale ends when the end date (21th March 2017, 4:00 pm GMT) is reached, or when the total amount of supplied Edgeless tokens (440 000 000 EDG) are sold.

If the Crowdsale campaign does not reach its capital goal of 50 000 ETH or the equivalent, all funds will be returned to the EDG holders by the Ethereum smart contracts.

5.1. Crowdsale summary

EDG created per 1 ether	1 000 EDG
Date of Crowdsale start	28th February 2017, 3:00 pm GMT
Date of Crowdsale end	21th March 2017, 4:00 pm GMT
Maximum number of EDG generated	500 000 000 EDG (unissued/ unsold tokens are automatically burned by a smart contract)
of which Crowdsale participants	440 000 000 EDG
of which the Edgeless team	50 000 000 EDG*
of which Crowdsale partners	10 000 000 EDG

*After the Crowdsale, Edgeless team's 50 000 000 tokens will be locked for 12 months by a smart contract.

5.2. Announcement of Crowdsale beginning

The Crowdsale beginning will be announced through the following channels:

- Project webpage: edgeless.io
- Official Twitter: twitter.com/edgelessproject
- Official Slack: edgelesscasino.slack.com
- Reddit: reddit.com/r/edgeless

5.3. Edgeless Tokens (EDG)

In a full Crowdsale Edgeless will issue 500,000,000 Edgeless tokens (EDG). Edgeless tokens will be an Ethereum-based token of value. The tokens will be used only to play in Edgeless casino.

Our Ethereum-based token contract complies with the ERC20 standard, which means that it is extremely easy for exchanges involving trading edgeless tokens to happen. More detailed info about the ERC20 standard: <https://github.com/ethereum/EIPs/issues/20>

5.4. Crowdsale “Power Hour” rewards

The first hour of the Crowdsale will be a Power Hour. During the Power Hour, 1 Ether will buy 1200 EDG. After that, the price will change to 1100:1, and then decrease every week until it reaches 1000:1 in the third week:

Power hour (28th February 2017, 3:00 pm GMT - 28th February 2017, 4:00 pm GMT):	1ETH = 1200 EDG
1 week (28th February 2017, 4pm GMT - 7th March 2017, 4pm GMT):	1ETH = 1100 EDG
2 week (7th March 2017, 4pm GMT - 14th March 2017, 4pm GMT):	1ETH = 1050 EDG
3 week (14th March 2017, 4pm GMT - 21th March 2017, 4pm GMT):	1ETH = 1000 EDG

5.5. Development Roadmap

1. Phase - 50 000 ETH

In the first phase, we will develop 0% house edge Blackjack, obtain a license and finalize the license.

2. Phase - 110 000 ETH

At this phase, most of the funds will be allocated towards security/hacker prevention and increased size of Edgeless Casino's bankroll so that it can payout any larger wins. A bigger bankroll means a larger betting range for visitors and more wagered money as a result.

3. Phase - 190 000 ETH

We will work on the development of a new game - 0% house edge video poker. It is based on luck and skill, the same as Blackjack. The second game is a micro limit dice - a pure luck game.

4 Phase - 250 000 ETH

This phase is all about the sports betting development. It has a smaller scale but better return on investment (the "ROI") than traditional games. Our Edgeless way will offer a much lower edge than other casinos. Sports betting will be so that our casino can stay competitive.

5 Phase - 360 000 ETH

The Grand Vision - a platform for developers where they can easily take our tools and develop their own fully transparent casinos so cryptocurrency and blockchain can become the standard of the casino industry. And later on, the standard of the world.

5.7 Timeline

Date of Crowdsale announcement	(14th December, 2016)
Date of Crowdsale starts	(28th February 2017)
Date of Crowdsale ends	(21th March 2017)
0% edge BlackJack Live	(Q4 2017)
Dice and other games Live	(Q1 2018)
Sports Betting Live	(Q3 2018)
Poker Live	(Q4 2018)

5.7. Crowdsale Ethereum-based trustless smart Contract key points:

/* From this part of contract EDG holder can see four main dates of crowdsale, and four different prices */

```
    address public beneficiary;
    uint public fundingGoal;
    uint public maxGoal;
    uint public amountRaised;
    uint public start;
    uint public tokensSold;
    uint[4] public deadlines = [1484020800, 1484625600,
1485230400,1485748800];
    uint[4] public prices =
[909090909090909,952380952380952,970873786407766, 10000000000000000];
    token public tokenReward;
    mapping(address => uint256) public balanceOf;
    bool fundingGoalReached = false;
    event GoalReached(address beneficiary, uint amountRaised);
    event FundTransfer(address backer, uint amount, bool isContribution);
    bool crowdsaleClosed = false;
```

/* Here is minimal and maximal token sale amount, if minimal amount will not be reached, sent ethers will be returned to Edgeless Crowdsale participants */

```
    function crowdsale( ) {
        beneficiary = 0x2aA2B0Ca9405B882e02851B81706904829C4AF17;
        fundingGoal = 50000000;//in tokens
        maxGoal = 394240000;//in tokens
        start = 1484146800;
        tokenReward = token(0xec1edef9c3c6035ee89589f516b20929a9225a24);
```

/* The function without name is the default function that is called whenever anyone sends funds to a contract, will receive Edgeless tokens */

```
    function () payable{
        uint amount = msg.value;
```

```
uintnumTokens = amount / getPrice();
if (crowdsaleClosed||now<start||tokensSold+numTokens>maxGoal) throw;
balanceOf[msg.sender] = amount;
amountRaised += amount;
tokensSold+=numTokens;
if(!tokenReward.transferFrom(beneficiary, msg.sender, numTokens)) throw;
FundTransfer(msg.sender, amount, true);
}
```

6. The Edgeless team

Ignas Mangevicius, Co - Founder, Blockchain-Based Systems Developer, Etherslots Founder, [Linkedin](#)

He is a father and a husband. He profoundly enjoys developing centralized and decentralized systems. He is a university educated Information Systems Engineer who has been working in IT for more than 10 years. He is now focused on smart contracts and Ethereum-based decentralized apps that he sincerely believes will make a better world. His most recent project is the [Etherslots.win](#), an app built on Ethereum blockchain.

Tomas Draksas, Co-founder, Professional Poker Player & Gambler, [Linkedin](#)

In 2008 - 2013, online poker was on the rise and a lot of new players were jumping into the game. At that time, Tomas decided to study the math of probability and poker. Later on, his direction changed to investing, cryptocurrency and digital marketing. He is the CEO of a conversion rate design company (Convi Design) which provides solutions to start-ups in the field of conversion rates and sales funnels. Additionally, he is investing into other start-ups. His key competences which are needed for Edgeless Casino are: probability math and models, gambling theory and psychology, digital marketing.

Tomas Lukosaitis, Co - Founder, Operations & Business, [Linkedin](#)

Tomas is coming from the field of P2P technologies, the sharing economy and FINTECH industry. He was working as digital marketer in a platform "SAVY". It is a peer to peer lending platform which connects investors to consumer loan offers. Currently Tomas owns an IT development company which is specialising in Mobile App Development. His key competences which are needed for Edgeless Casino are: digital marketing, operations management.

7. Blockchain developers

Dipl.-Ing. Julia Altenried, Smart Contract and Dapp Developers

She studied computer science at the University of Salzburg and the Politechnical University of Madrid. Julia has gathered over 8 years of experience as a software engineer, working for different international companies, and then founded JAM Data. Julia specialized in IT-Security and later blockchain technology with a focus on Ethereum smart contract development. In Blockchain technology, she sees the potential to increase transparency in governmental operations as well as business processes, thereby reducing corruption worldwide. When not working, she is traveling around the globe or practicing Yoga.

Dipl.-Ing. Stefan Höller, Smart Contract and Dapp Developers

Stefan was formally educated at the University of Salzburg. His career started 15 years ago as the e-commerce department head for Austria's largest bicycle retailer. He's a professional mean-fullstack engineer and early blockchain enthusiast. Currently, he is consulting for companies regarding decentralized applications. Stefan sees blockchain technology as a powerful tool to bring equal chances to developing countries. He is a fitness-freak and passionate about exploring foreign cultures.

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